

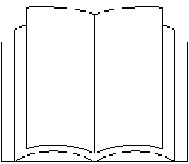


Operator's manual for Attachment



Concrete Mixing Bucket

Product number: 500-700-series loaders: A21406



Please read this manual carefully before using the equipment, and follow all instructions.

Keep this manual for later reference.

Manufacturer:

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Appendix: EC Declaration of Conformity

This manual uses the following warning symbols to indicate factors that must be taken into account to reduce the risk of personal injury or damage to property:

	This safety symbol refers to important safety information in this manual. It warns of an immediate hazard that could cause serious personal injury or damage to property. Read the warning text accompanying the symbol carefully and ensure that other operators are also familiar with the warnings, since personal safety is at stake.
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This notice symbol indicates information about the correct operation and maintenance of the equipment.

Failure to observe the instructions accompanying the symbol can lead to equipment breakdown or other property damage.

In addition to the safety instructions included in this manual, you must observe all occupational safety regulations, local laws and other regulations concerning the use of the equipment. Due to continuous product development, some of the details in this manual may differ from your equipment. We reserve the right to change the contents of the manual without notification. This operator's manual is a translation of the original Finnish version.

1. Foreword

Avant Tecno Oy would like to thank you for your purchase of this AVANT attachment. It has been designed and manufactured on the basis of years of product development and experience. By familiarising yourself with this manual and following the instructions, you ascertain your safety and ensure the reliable operation and long service life of the equipment. Read the instructions carefully before starting to use the equipment or performing maintenance.

With these instructions, even users with only little experience can use the concrete mixer and loader safely. The manual also includes important instructions for experienced AVANT operators. Ensure that all persons using the equipment have received proper guidance and familiarised themselves with the manual and all safety instructions before using the equipment. Keep this manual at hand throughout the service life of the equipment. If the manual is lost, you can request a new one from the retailer or manufacturer. If you sell or transfer the equipment, be sure to hand over this manual to the new owner.

2. Designed purpose of use

This AVANT Concrete mixer is an attachment designed and manufactured to suite AVANT multi purpose loaders for mixing and easy dispensing of concrete. With the mixer concrete can be prepared and transported easily even at difficult locations. Concrete mixer is hydraulically operated and it can be emptied by tilting the bucket or by opening the ejecting hatch at the side of the bucket and using the chute to channel the concrete accurately.

Concrete mixing bucket is a handy tool when you need to make concrete without electricity or when concrete needs to be transported to a difficult place. Sand or gravel can be filled directly from the ground by using the mixer like a normal bucket. The bucket can also be filled manually, when the bucket is supported against the ground. The attachment is suitable for handling concrete or other similar type of material.

The attachments have been designed to require as little maintenance as possible. Regular maintenance tasks can be performed by the operator, but certain more demanding tasks must be left to qualified professionals. Appropriate protective equipment and original spare parts must be used for the maintenance. Read the maintenance and servicing instructions in this manual. Please contact your AVANT retailer if you have additional questions about the operation or maintenance of the equipment or if you require spare parts or maintenance services.

3. Safety instructions for the concrete mixer

Please bear in mind that safety is the result of several factors. The loader-attachment combination is highly powerful, and improper or careless use may cause serious personal injury or property damage. Due to this, you should carefully familiarise yourself with the manuals of both the loader and the attachment before starting operation. Do not use the attachment if you have not familiarised yourself with its operation and the related hazards. The purpose of these instructions is to help you to:

- use the equipment in a safe and efficient manner
- monitor and prevent any hazardous situations
- keep the equipment intact and ensure a long service life



- When attaching the attachment to the loader, ensure **that the locking pins of the loader's quick attach plate are in the lower position and that they have locked the concrete mixer to the loader.**
- The loader and its controls are to be used by only one operator at a time. Do not go near the attachment if another person is operating the controls of the loader, or allow others to go near the machine when operating the loader.
- Avoid the hazard of falling materials by keeping the bucket always close to the ground. Be sure not to tilt the concrete mixer too far up to prevent material from spilling from the bucket.



- Always transport the mixer as low and as close to the machine as possible to keep the centre of gravity low.
- When filling the concrete mixer as a normal concrete mixer, keep it securely against the ground and attached to the loader.
- Watch out for the hazard getting entangled around the mixing shaft. Always keep the safety grille in place when operating the mixer.
- Before starting the concrete mixer, make sure that there is no cured concrete or other foreign objects in the bucket that could get ejected from the attachment or that could damage the mixer.
- Shut down the loader engine before any maintenance or adjustments. Ensure that the attachment is properly supported during maintenance, inspections or cleaning.
- Never go under a raised concrete mixer or the loader boom.
- When driving with the bucket attached, pay attention to the contours of the terrain and other hazards, such as branches and trees reaching the driver's seat. Pay attention to the surroundings and any other persons and machines moving in the vicinity.
- When taking the concrete mixer off from the loader, ensure that it is secured against accidental movement. To prevent the attachment from tipping over, lower it on even surface. Ensure that the concrete mixer is supported properly during storage periods.
- Do not modify the structure of the attachment in a manner that would affect its safety. Use the concrete mixer for its intended use only.
- Check the mixing tools regularly. Read the instructions on maintenance and servicing and only use original spare parts.
- Also read the safety instructions for the loader in the loader manual.



Watch out for crushing hazard under the loader boom or crushing under or in front of the attachment. Note that the loader boom moves down and may be tilted even if the engine has been stopped. Never go under a raised boom or to the danger area between the boom and the attachment. Avoid staying directly in front of the mixer.



Improper or careless use of the mixer may cause hazardous situations. Use the mixer only for preparing concrete or plaster from raw materials suitable for this purpose. Familiarise yourself with the loader controls in a safe area. Pay particular attention to the safe stopping of the attachment and the machine.



Never leave the machine unattended while the engine is running. Prevent unauthorized use of the equipment.



Note the effects of a heavy load to the handling characteristics of the loader. Estimate the load beforehand; typical density of concrete is around 2400 kg/m³. When driving, keep the telescopic boom retracted and keep the load close to the ground. Use additional counterweights as needed. Read the operator's manual of the loader.





When working at construction sites, in many cases a falling objects protective structure (FOPS) needs to be installed on the loader. Find out about necessary safety equipment. Contact your retailer for availability and installation of the protective roof.



Remember to wear proper protective clothing:



Never wear loose clothing, long scarves, necklaces, jewellery or other items that might become entangled with the rotating parts of the concrete mixer. If needed, wear a hat to prevent long hair from coming into contact with the rotating parts.



- The noise level in the driver's seat may exceed 85 dB(A). Wear hearing protection while working with the loader.



- The use of safety boots is recommended while working with the loader.
- Wear protective gloves when handling hydraulic hoses.



- Wear a dust mask when loading cement or operating in dusty conditions.

4. Concrete mixer specifications

Table 1 –Concrete mixer technical data

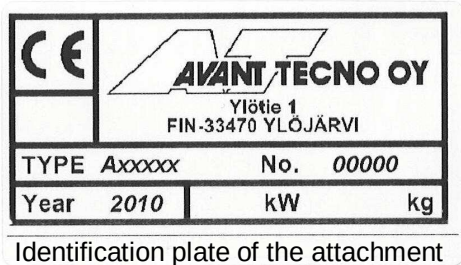
Product number	A21406
Total width:	1 590 mm
Volume:	270 l
Weight:	490 kg

Maximum input of hydraulic energy:	70 l/min 22,5 MPa (225 Bar)
Recommended operating speed:	0,3 – 0,8 rev/s (25-70 l/min)
Suitable AVANT loader:	(520) 528–700

Listed below are the markings and labels fastened to the flail mower, which must be visible on the equipment. Replace the warning labels, if they have become unclear or detached completely. New labels are available via your retailer or contact information provided on the cover page.



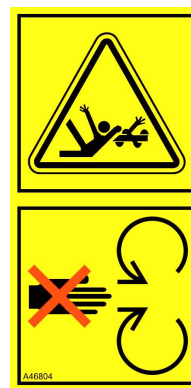
The warning decals contain important safety information. Replace damaged or missing decals with new ones.



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A46772



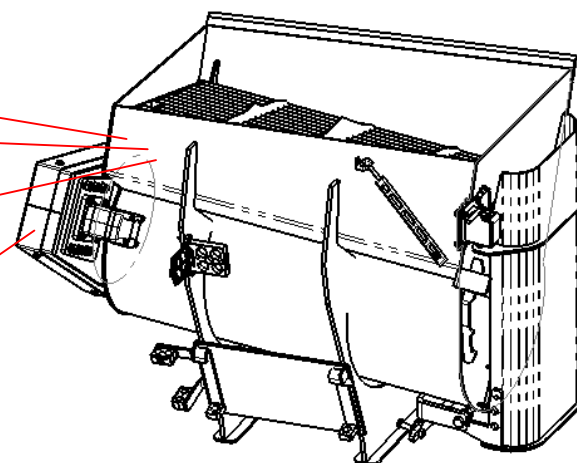
A46804



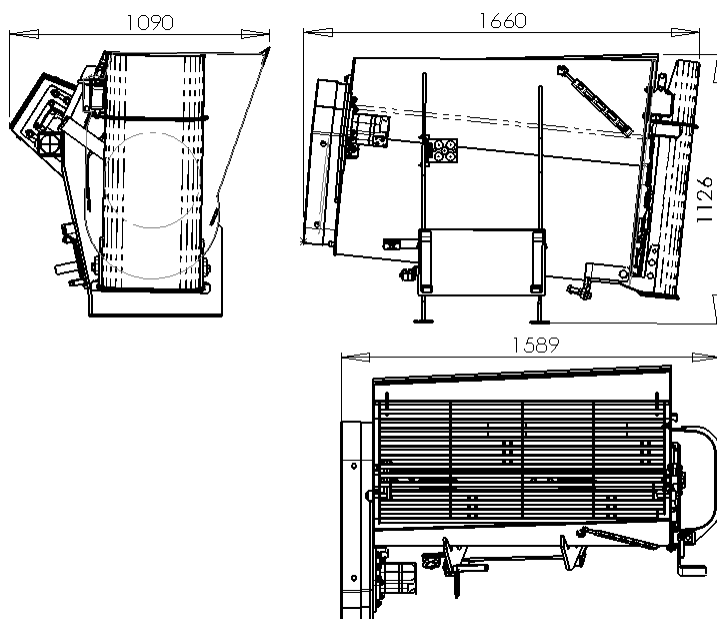
A46798

Table 2 – Decal warning messages

Decal	Explanation
A46771	Read instructions before use.
A46772	Do not go under a raised attachment or the loader boom, keep a safe distance to lifted equipment.
A46804	Hazard of entanglement, keep clear of moving parts. Keep the protective grille attached.
A46798	Crushing and entanglement hazards between transmission components. Chain drive, do not operate without cover in place.



Warning decal locations



Dimensions in operation and transportaion position

5. Assembly of the concrete mixer

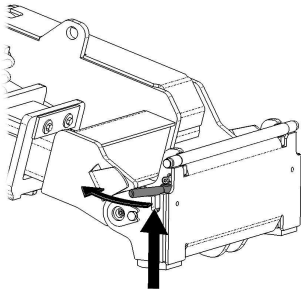
The concrete mixers are delivered in working condition. Attach the attachment and its hydraulic hoses according to the following instructions and choose a safe working area. Read the instructions in this manual for mounting, using and maintenance of the concrete mixer before operating the equipment.

5.1. Attaching the concrete mixer bucket to the loader

Attaching the attachment to the loader is quick and easy, but it must be done carefully. If the attachment is not locked to the loader, it may detach from the loader and cause hazardous situations. In the worst case, the attachment may slide onto the driver along a raised boom. For this reason, the attachment must under no circumstances be raised over one metre when it has not been locked. To prevent hazardous situations, always follow the attachment mounting instructions provided in the following pages. Also remember the safety instructions from Chapter 3.

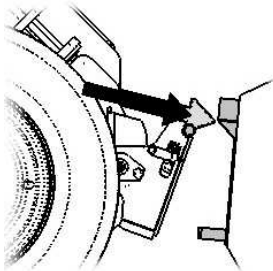
The attachment is installed to the loader boom by using the quick attach plate and the counterpart on the attachment. The locking on the loaders is semi-automatic, but due to position of the coupling plate on the concrete mixer, we recommend to lock the locking pins manually at the same time as the hydraulic hoses are connected. The concrete mixer is attached to the loader as follows:

Step 1:



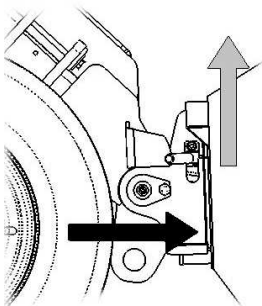
- Lift the quick attach plate locking pins up and turn them backwards into the slot so that they are locked in the upper position.
- If your loader is equipped with a hydraulic attachment locking system, see additional instructions about the use of the locking system from the relevant manuals.
- Ensure that the hydraulic hoses are not in the way during installation.

Step 2:



- Turn the quick attach plate hydraulically to an obliquely forward position.
- Drive the loader onto the attachment. If your loader is equipped with a telescopic boom, you can utilise this.

Step 3:



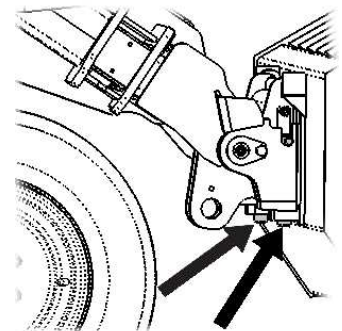
- Lift the boom slightly – pull the boom control lever backward to raise the attachment off the ground.
- Turn the boom control lever left to turn the bottom section of the quick attach plate onto the attachment.
- Lock the locking pins manually.



Unlocked attachment may fall on the boom or operator and cause serious injuries. Before moving or lifting the attachment, ensure that the locking pins are in the lower position and come through the fasteners on both sides.



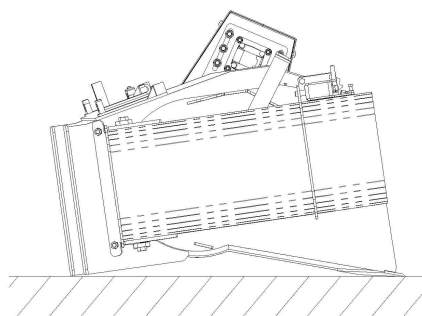
Do not use the automatic return of the locking pins when the attachment or the boom has been lifted **at a height of over half a metre**. Excess tilting increases the risk of toppling over. We recommend that you lower the attachment onto the ground and secure the locking manually.



5.2. Storage of the bucket

When finished working with the concrete mixer and it has been thoroughly cleaned after use, lower it on solid and even surface, where it cannot move by accident. To eliminate the risk of the attachment from tipping over, lower it as shown in the adjacent figure.

Read instructions shown in chapter 7 and service the bucket before long-term storage. To avoid rust damage, do not store the attachment directly against the ground and do not allow water to stay in the bucket.



Storage position of the bucket



Prevent overturning of a stored attachment

5.3. Connecting and disconnecting the hydraulic couplings

The attachment is equipped with either standard quick couplings or the multiconnector system. If you want to change the type of the hydraulic couplings, contact your Avant dealer or service point for instructions or installation services.



Never connect or disconnect quick couplings or other hydraulic components while the system is under pressure or if the control lever of the auxiliary hydraulics control lever is locked on. Connecting or disconnecting the hydraulic couplings while the system is pressurized may lead to unintended movements of the attachment, or ejection of high-pressure fluid, which can cause serious injuries.

Keep the fittings as clean as possible; use the protective caps for the quick couplings on both the attachment and the loader. Dirt, ice, etc. may make using the fittings significantly more difficult. Never leave the hoses hanging on the ground; place them on top of the equipment.

Connecting standard quick couplings:

Before connecting or disconnecting the standard quick couplings, shut down the loader engine and move the auxiliary hydraulics lever on the loader to its extreme positions a few times. This way, there will be no residual pressure in the system and it is possible to connect the fittings.

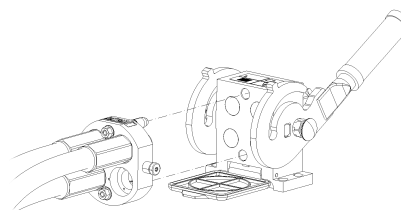
To connect and disconnect the standard couplings, move the collar at the end of the female fitting. The hoses should be connected so that the coupling equipped with a coloured cap is connected to the corresponding coupling of the loader. Note that the protective caps on the loader and the attachment can be fastened to each other during operation to reduce the accumulation of dirt. When disconnecting the standard quick couplings a small amount of oil may drip from the couplings. Wear protective gloves and have some cloth at hand to keep the equipment clean.



Connecting the multiconnector system:

Align the pins of the attachment connector with corresponding holes of the loader connector. The multiconnector will not connect if the attachment connector is upside down. Connect and lock the multiconnector by turning the lever towards the loader.

The lever should move easily all the way to its locking position. If the lever does not slide smoothly, check the alignment and position of the connector and clean the connectors. Also shut down the loader and release the residual hydraulic pressure.



Disconnecting the hydraulics:

Before disconnecting the fittings, lower the attachment on solid and level surface. Turn the control lever of the auxiliary hydraulics to its neutral position.



When removing the attachment, always disconnect the hydraulic couplings before unlocking the quick attach plate, to prevent hose damage and any oil spills. Reinstall the protective caps on the fittings to prevent impurities from entering the hydraulic system.

Releasing residual hydraulic pressure:

In case residual pressure is left in the hydraulic system of the attachment, it is often possible to disconnect the hydraulic couplings, but it may be difficult to connect them the next time. If the fittings will not connect, the residual pressure must be released by turning the auxiliary hydraulics control lever of the loader, when the engine is turned off. To make sure that there will not be residual pressure in the hydraulic system of the attachment, shut down the loader engine and move the auxiliary hydraulics control lever of the loader before disconnecting the couplings.

6. Instructions for use

Once the concrete mixer bucket and its hydraulic hoses have been attached according to the instructions, it is ready for work. Read the following instructions about work site, setting up, use and maintenance of the concrete mixer.



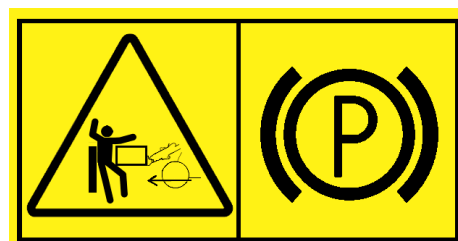
Never go under a raised boom or concrete mixer component. Note that the loader boom can be moved even if the engine has been stopped.

Check the attachment and the operating environment once more before starting the machine. Quick inspection of the concrete mixer and the operating environment prior to use are a part of ensuring safety and the best performance of the equipment.

- Choose a level working site, which can carry the load of the equipment.
- Operate the concrete mixer in daylight or in good artificial light and in well ventilated areas. All obstacles around the bucket must be removed. Remember correct working methods and avoid getting down unnecessarily from the loader.
- Transport the mixer carefully and note the effect of weight at the front to the stability of the loader.
- Make sure that the parking brake of the loader is engaged and that the attachment is supported on the ground, so that it can not move when loading material by hand.
- Check that the concrete mixer is intact and remove any material left in the bucket before starting the mixer. Check that the mixing shaft can rotate freely and that there is no material coiled around the shaft.
- Check regularly to ensure there are no hydraulic leaks. Refer to chapter 7 for maintenance instructions.
- Ensure that the loader and the attachment are used in a safe and intended manner. Do not allow children to operate the equipment.
- Do not go near the equipment when another person is operating the loader controls.

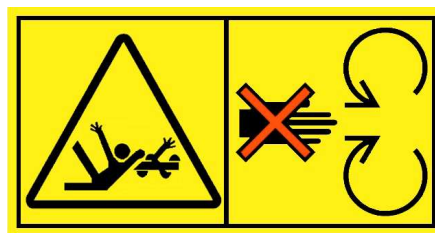


Avoid hazards of crushing and being overrun by ensuring that the loader parking brake is engaged. Use the equipment only on level surfaces and make sure that the equipment will stay still regardless of the possible movements caused by the mixer.





Beware of the rotating mixing shaft of the mixer, which causes a hazard of entanglement. Keep a safe distance to the mixer when it is in operation. Never reach into the rotating mixer with any means. Also, never leave the mixer running unintended. Shut down the loader engine and prevent unintended starting, if reaching to the mixing drum is necessary. Always keep the safety grille attached.



Check before use that all protective covers are in place. Do not use the attachment, if the safety grille in the bucket or the transmission cover at the side of the bucket are not in place. Never perform any adjustments, maintenance or cleaning of the equipment if the mixer can start by accident. Shut down the loader engine first.



Overheating of hydraulic oil decreases power and affects the service life of the hydraulic components of both the attachment and the loader. Clean the loader oil cooler core regularly; refer to the loader manual for more instructions.

Filling and using the bucket:

The concrete mixer bucket can be filled from a pile like a normal bucket. Do not rotate the mixing tools while loading the mixer as a normal bucket. When filling the bucket in a similar way as a regular concrete mixer, keep the rotating speed of the mixing shaft low to avoid ejection of materials.

The concrete mixer is started by turning the auxiliary hydraulics control lever to its locked position. Usually low engine rpm and 1-pump setting of the auxiliary hydraulics are enough. Depending on the operating conditions, the operating speed can be increased after the bucket is filled, when risks of falling or thrown materials have decreased.



Note that the mixer may eject gravel or other material when operated at high rotating speeds. Keep the operating speed reasonable especially at the beginning of mixing; use the 1-pump setting of the auxiliary hydraulics.

The bucket can be emptied with the discharge opening at the side of the bucket or by tilting the bucket as a regular bucket. Lower the rotating speed and set the discharge chute before opening the discharge hatch. To assist the pouring of the concrete, rotate the mixer to opposite directions. Start pouring the concrete carefully. The discharge chute angle can be adjusted with the two screws near the discharge chute.



Reaching to the mixer through the pouring hatch may lead to severe injuries. Stop the mixer before attempting to scrape the last of the concrete in the bucket. Never reach to the bucket while the mixer is in operation and keep feet away from the hatch.

7. Inspections, maintenance and servicing

The most important factor in continuous maintenance of the concrete mixer is thorough and prompt cleaning of the equipment. Maintenance includes monitoring the condition of the mixing tools as well as regular lubrication of the equipment. The drive chain must be checked and adjusted at least annually or after every 50 hours of use.



Ensure that the concrete mixer is properly supported during all cleaning, inspection and maintenance procedures. Lower the attachment on even ground and keep it attached on the loader. All maintenance procedures must be performed with the loader engine shut down and the hydraulic hoses disconnected.

Check the condition of the hydraulic hoses and components when the engine has been turned off and the pressure has been relieved. Repair all leaks immediately after detecting them; a small leak can quickly grow into a big one. Do not use the equipment, if you have discovered a fault in the hydraulic system. Leaking hydraulic fluid may penetrate the skin and cause serious injuries. Hydraulic fluid is also harmful to the environment.

Check hoses visually for cracks or abrasions. Monitor the wearing of the hoses and stop the use, if the surface layer of the hose has worn off. If there are signs of leaking, to check the component, hold up a piece of cardboard in the area where a leak is suspected. Finding any of these faults means that the hydraulic hose or component in question must be replaced. Spare parts are available from your nearest AVANT retailer or service point. Leave the service for professional maintenance, if unsure how to perform the repair safely.



Never tighten a hydraulic fitting under pressure, since the fitting may break and the released oil may cause serious injuries. Do not use the equipment, if you have discovered a fault in the hydraulic system.

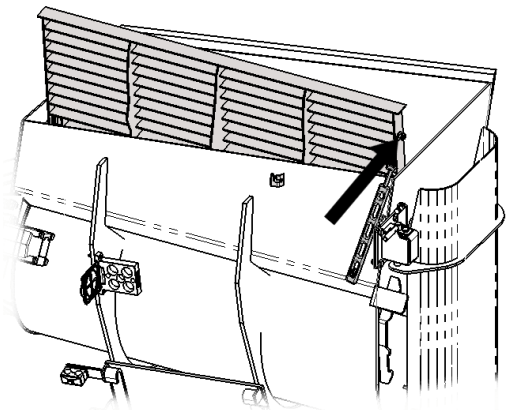


7.1. Cleaning the equipment

Clean the concrete mixer immediately after use, since this makes cleaning easier. Unload any remaining material from the bucket to a suitable place. Use only water to clean the equipment. You may use a pressure washer and a mild detergent. Do not use strong solvents or spray directly at the hydraulic components or at the labels on the attachment either.

Touch up the paint where needed, or oil the metal parts lightly, and grease the lubrication points to prevent rust damage. For storage, place the mixer bucket to the position shown in section 5.2. Do not store the attachment directly against the ground; place it on blocks of wood or on a pallet, for example.

To make cleaning of the mixer easier, the protective grille can be tilted and locked to upright position with the rubber strap shown in adjacent figure. The grille is intended to protect the operator and it must be placed to its operating position whenever operating the mixer. The hazard of entanglement or crushing persists during maintenance, do not operate the machine when the grille is lifted.



7.2. Lubricating the mixing bucket

The bucket has two lubrication points, to which a small amount of lubricant is added every 5-10 hours of use. The lubrication points are located at the bearings at both ends of the mixing shaft. The lubricating point at the free end of the axle is directly accessible. To lubricate the transmission end of the axle, the chain housing cover must be removed. The lubrication nipples are standard R1/8" nipples. Replace any damaged nipples. Clean the chain housing cover when removing it and lubricate the chain with suitable oil. Check the chain as shown in Section 7.4.



Lubricant must be added approximately every 5 to 10 operating hours. Clean the end of the nipple before greasing. Only add the grease amount of a couple of grease press pushes at a time.

7.3. Adjusting the mixing tools

The concrete mixer has a total of 8 mixing pads. Adjust the pads to a distance of about 5 mm from the mixer walls. The spare part number for a replaceable rubber wear pad is A412429.

7.4. Checking and adjusting the drive chain

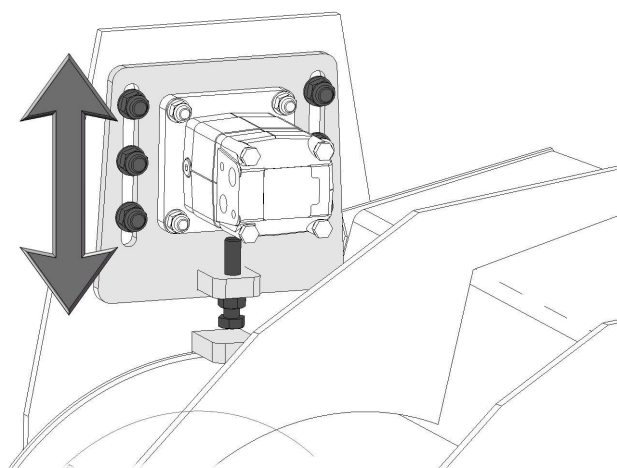
Checking the condition and the tension of the chain must be done at least annually. To check and adjust the chain, place the bucket on a stable and even platform, shut down the loader engine and disconnect the hydraulic couplings. Make sure that during the work the bucket is supported so that it can not move at all. Any movements of the mixing tools create a pinching hazard for hands between the chain and the sprockets.

To check the roller chain, remove the protective housing and press the chain at its longest section, between the sprockets. The clearance must be about 0.5 - 1 cm.

If necessary, tension the chain by moving the hydraulic motor. Loosen the six screws on the hydraulic motor base. After this, move the base with the tightening screw under the motor, until the tension is correct.

Tighten the loosened base screws and recheck the tension of the chain once more, as it may have been influenced by tightening the hydraulic motor base.

When the chain is set correctly, lubricate the chain and reattach the protective housing.



Tension the chain correctly. Over tightening of the chain damages the hydraulic motor and the bearings.

8. Warranty terms

Avant Tecno Oy grants a warranty of one year (12 months) for the attachment from the date of purchase.

The warranty covers repair costs as follows:

- Work costs are covered, if the repair is not performed at the factory.
- The factory replaces any defective components or consumables.

The factory may reimburse the price of components purchased by the customer in special cases that have been agreed in advance.

The warranty does not cover:

- Normal maintenance work, such as the replacement of blades or the belts, or parts and consumables required for it.
- Damages caused by unusual operating conditions or ways of use, negligence, structural changes made without the consent of Avant Tecno Oy, use of non-original parts or lack of maintenance.
- Consequences of a defect, such as interruption of work or other possible additional damages.
- Travel and/or freight costs caused by the repair.

EY-vaatimustenmukaisuusvakuutus
EG-försäkran om överensstämmelse
EC Declaration of Conformity
EG-Konformitätserklärung



Valmistaja / Tillverkare / Manufacturer / Hersteller:

AVANT TECNO OY

Osoite / Adress / Address / Adresse:

Ylötie 1

33470 YLÖJÄRVI, FINLAND

Vakuutamme täten, että alla mainitut tuotteet täyttävät konedirektiivin turvallisuus- ja terveysvaatimukset (direktiivi 2006/42/EY muutoksineen). Seuraavia yhdenmukaistettuja standardeja on sovellettu:
 SFS-EN ISO 12100, SFS-EN 982, SFS-EN 12151

Vi försäkrar härmed att nedan beskrivna produkter överensstämmer med hälso- och säkerhetskrav i EG-maskindirektiv (EG-direktiv 2006/42/EG som ändrat). Följande harmoniserade standarder har tillämpats:
 SFS-EN ISO 12100, SFS-EN 982, SFS-EN 12151

We hereby declare that the products listed below are in conformity with the provisions of the Machinery Directive (directive 2006/42/EC as amended). The following harmonized standards have been applied:
 SFS-EN ISO 12100, SFS-EN 982, SFS-EN 12151

Wir erklären hiermit, dass die nachstehend aufgeführte Maschine konform ist mit den Bestimmungen der EG-Maschinenrichtlinie (EG-Richtlinie 2006/42/EG mit Änderungen). Die folgende harmonisierte Normen wurden angewandt:

SFS-EN ISO 12100, SFS-EN 982, SFS-EN 12151

Kuvaus: Avant-kuormainten kanssa käytettävä betoninsekoituslaite
 Beskrivning: Betongblandare: arbetsredskap för Avant lastare
 Description: Concrete mixer; attachment for Avant loaders
 Beschreibung: Betonmischer; Anbaugerät für Avant Radlader

Mallit / Modeller / Models / Modellen:

Avant	
Betonikauha, 270 l Betongblandaeskopa, 270 l Concrete mixing bucket, 270 l Betonmischschaufel, 270 l	A21406

Allekirjoitus / Namnteckning / Signature / Unterschrift:

10.2.2011 Ylöjärvi, Finland

Risto Käkelä,
 Toimitusjohtaja / Verkställande direktör /
 Managing Director / Geschäftsführer





AVANT[®]